ROSE5730 Autonomous Mobile Robots

Lecture 2: Locomotion

Dongkun Han
Department of Mechanical and Automation Engineering;
The Chinese University of Hong Kong.



Outline

Locomotion

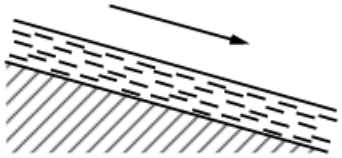
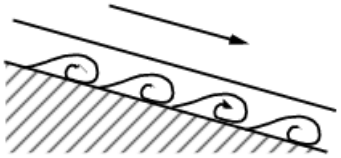
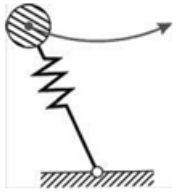
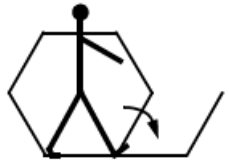
3 Wheeled locomotion

2 Legged locomotion

1 Key concepts



Locomotion on ground

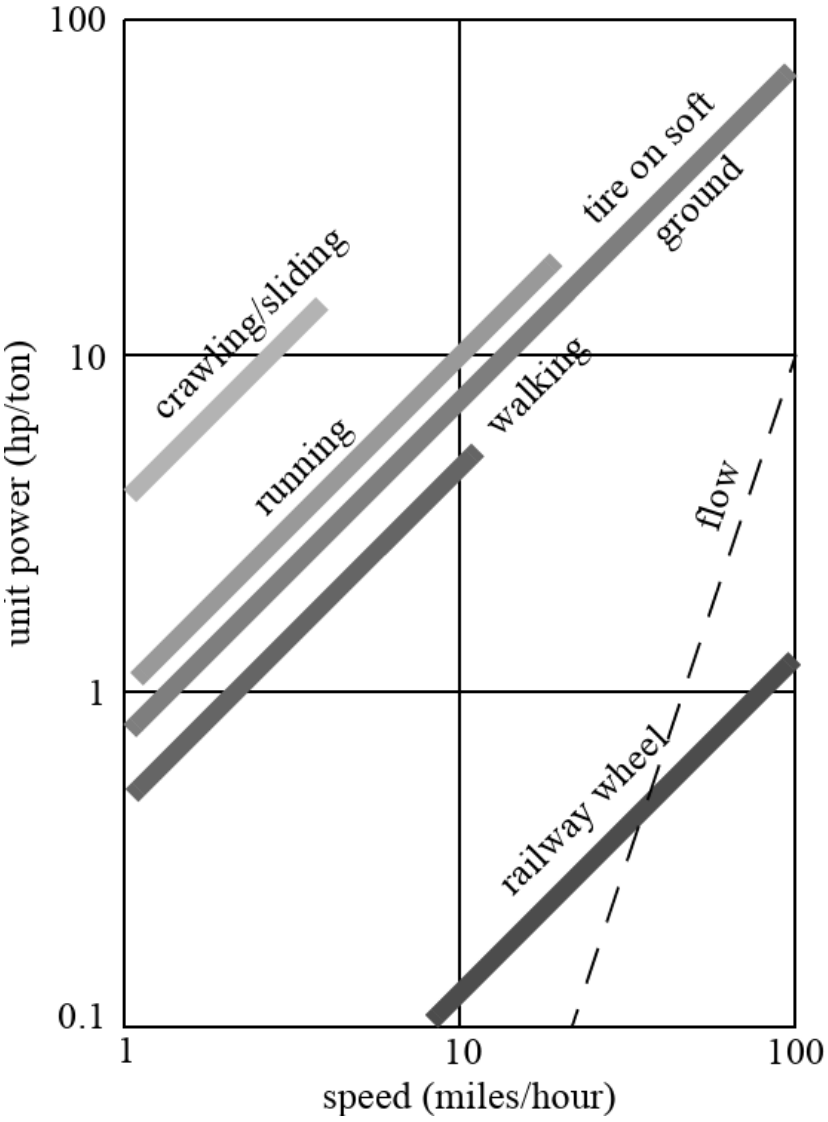
Type of motion	Resistance to motion	Basic kinematics of motion
Flow in a Channel 	Hydrodynamic forces	Eddies 
Crawl 	Friction forces	Longitudinal vibration 
Sliding 	Friction forces	Transverse vibration 
Running 	Loss of kinetic energy	Periodic bouncing on a spring 
Walking 	Loss of kinetic energy	Rolling of a polygon 

Locomotion concepts

- Nature came up with a multitude of locomotion concepts
  - Adaptation to environmental characteristics
  - Adaptation to the perceived environment (e.g. size)
- Concepts found in nature
  - Difficult to imitate technically
  - Do not employ wheels
  - Sometimes imitate wheels (bipedal walking)
  - The smaller living creatures are, the more likely they fly
- Most technical systems today use wheels or caterpillars. Legged locomotion is still mostly a research topic.

Why walking or rolling

- number of actuators
- structural complexity
- control expense
- energy efficient
  - terrain (flat ground, soft ground, climbing..)
- movement of the involved masses
  - walking / running includes up and down movement of COG
  - some extra losses



Characterization of locomotion

- Locomotion depends on the physical interaction between the vehicle and its environment.
- Locomotion is concerned with interaction forces, and the mechanisms and actuators that generate them.



T8X Robugtix robotic spider



Hexapod robot “RHex”

Issues of locomotion



UNIST Snake Robot

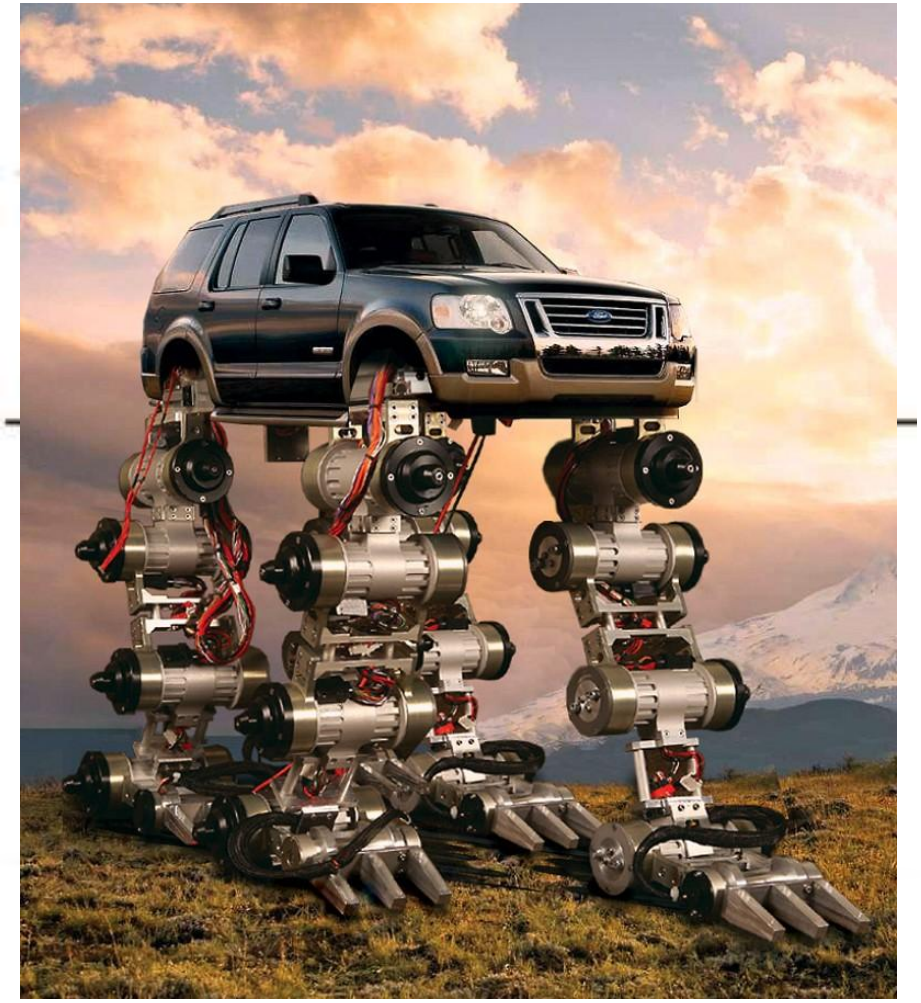
- stability
  - number of contact points
  - center of gravity
  - static/dynamic stabilization
  - inclination of terrain
- characteristics of contact
  - contact point or contact area
  - angle of contact
  - friction
- type of environment
  - structure
  - medium (water, air, soft or hard ground)

Legged robot

- **Wheels** are widely used in robotic systems.

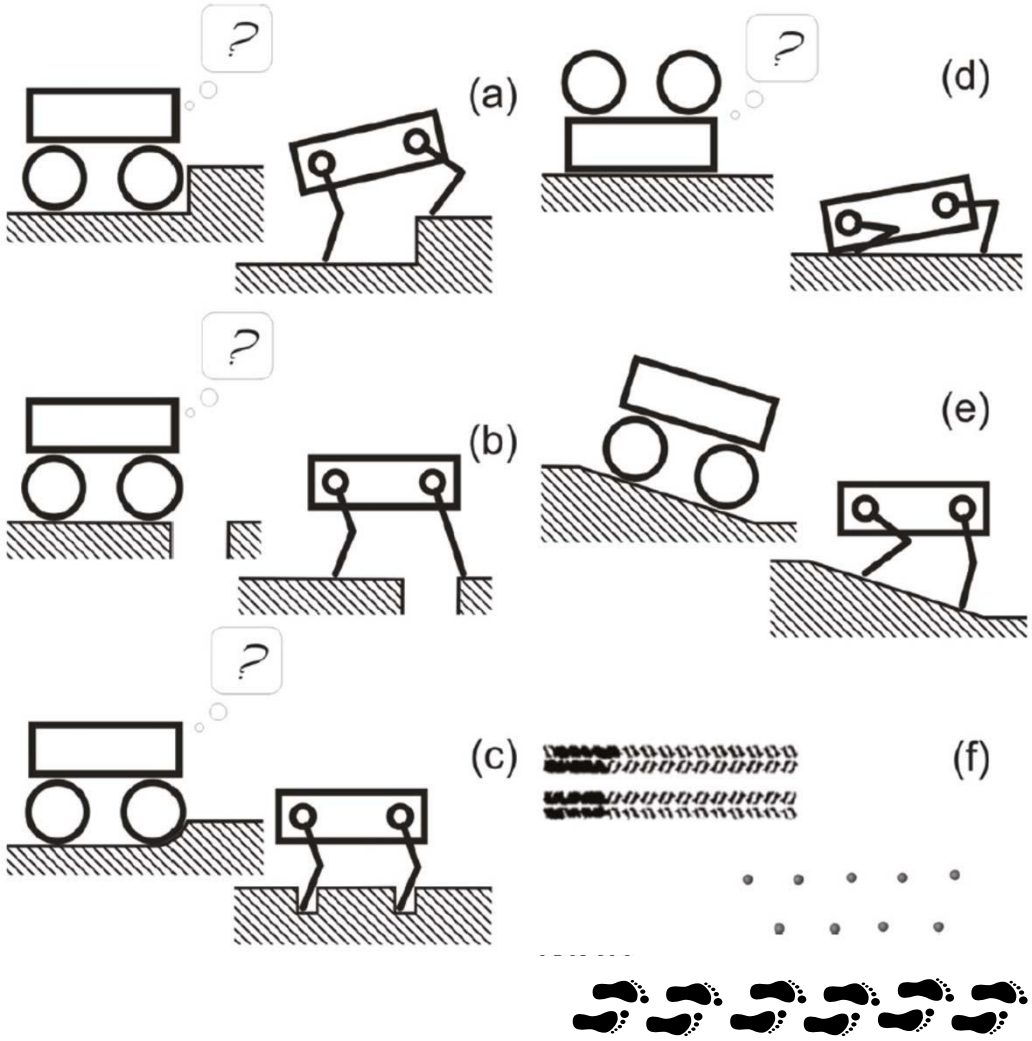






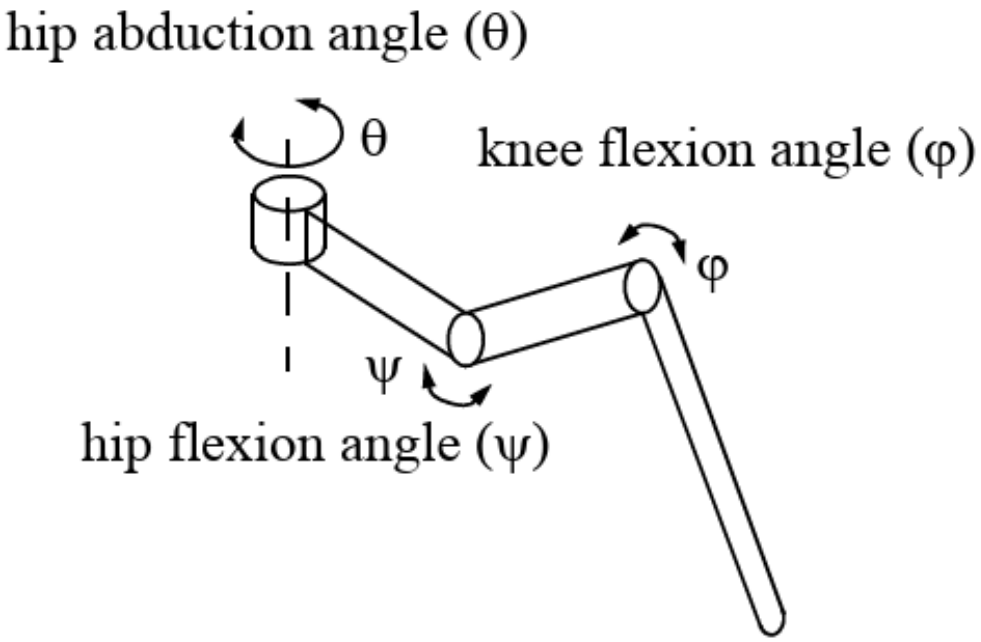
Why legged robots?

- Legged systems can overcome many obstacles, that are not reachable by wheeled systems.
- But it is quite hard to achieve since
  - many DOFs must be controlled in a coordinated way
  - the robot must see detailed elements of the terrain



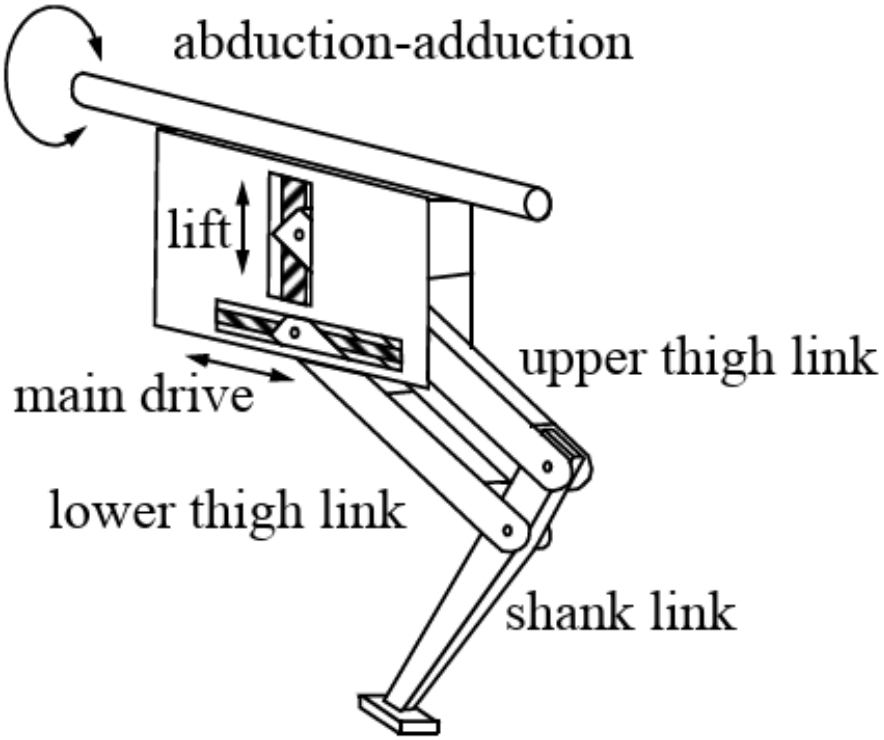
DOF of each leg

- A minimum of two DOF is required to move a leg forward
  - a lift and a swing motion.
  - Sliding-free motion in more than one direction not possible
- Three DOF for each leg in most cases



DOF of each leg

- 4th DOF for the ankle joint
  - might improve walking and stability
  - additional joint (DOF) increases the complexity of the design and especially of the locomotion control.



Gait

- The **gait** is a sequence of lift and release events for the individual legs.
  - It depends on the number of legs.
  - the number of possible events N for a walking machine with k legs is

$$N = (2k - 1)!$$

- For a robot with 6 legs (hexapod) N is already

$$N = 11! = 39{,}916{,}800$$

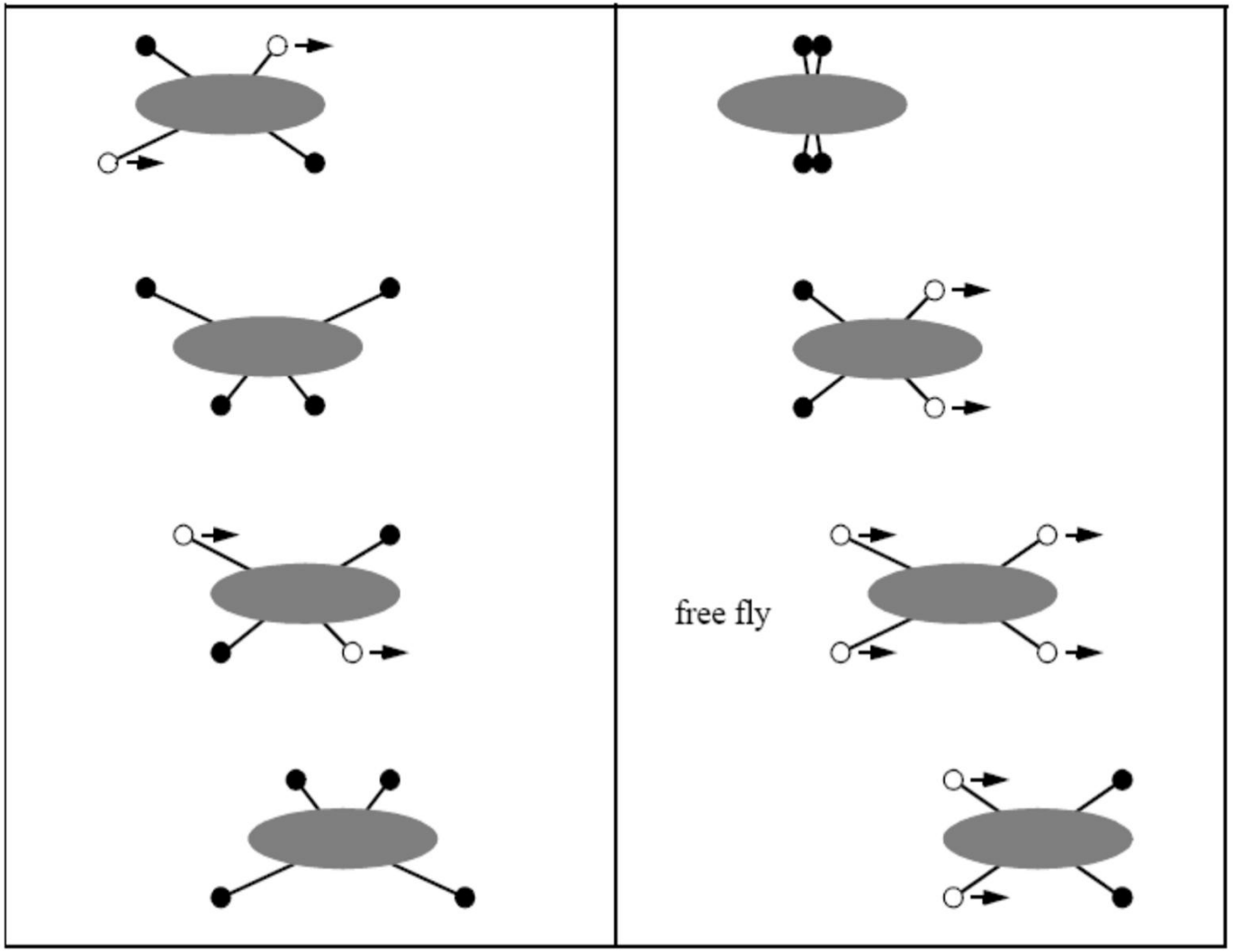
Gait

- For a biped walker ($k = 2$) the number of possible events N is

$$N = (2k - 1)! = 3! = 6$$

- The six distinct event sequences are:
  - 1. both legs down – right down / left up – both legs down;
  - 2. both legs down – right leg up / left leg down – both legs down;
  - 3. both legs down – both legs up – both legs down;
  - 4. right leg down / left leg up – right leg up / left leg down – right leg down / left leg up;
  - 5. right leg down / left leg up – both legs up – right leg down / left leg up;
  - 6. right leg up / left leg down – both legs up – right leg up / left leg down.

Common gaits with 4 legs

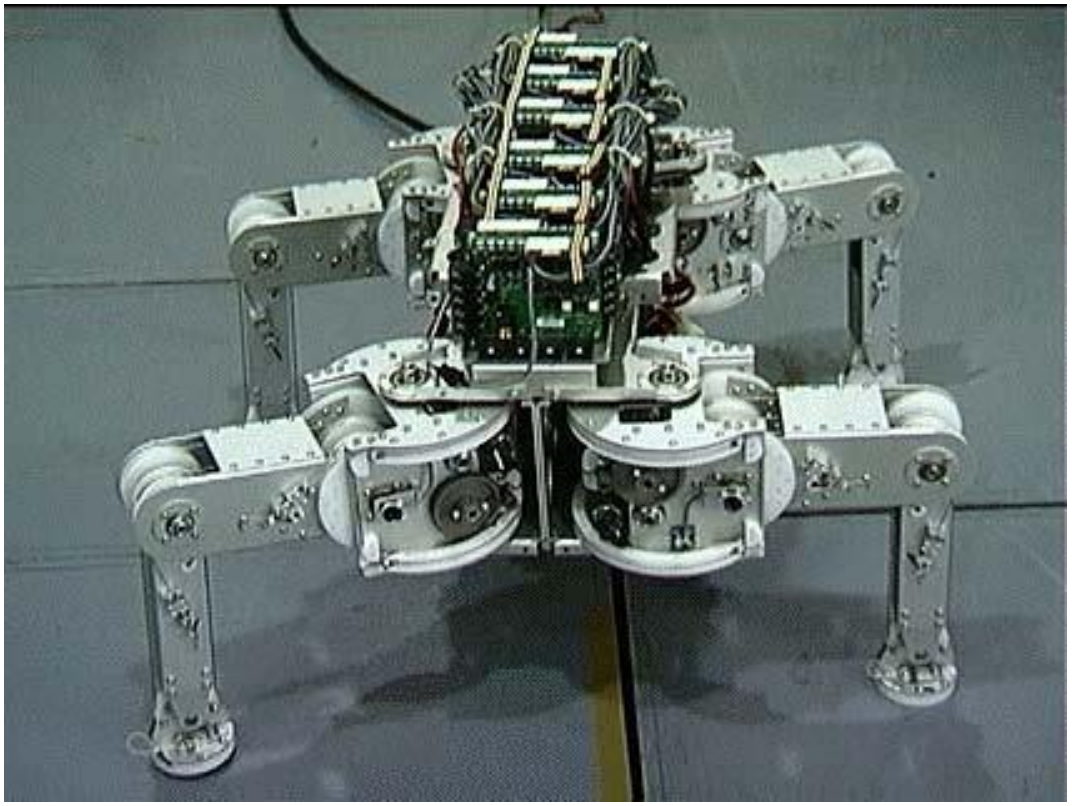


Changeover walking

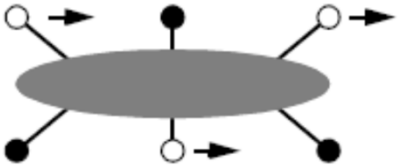
Galloping

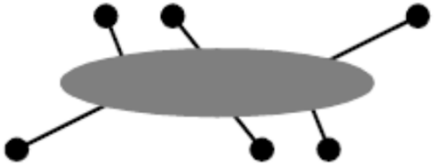
Quadruped walking robots

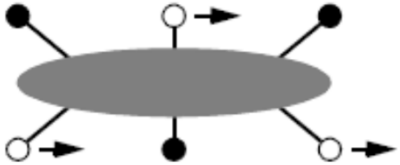
- Titan VIII, a quadruped robot, Tokyo Institute of Technology
  - Weight: 19 kg
  - Height: 0.25 m
  - DOF: 4*3

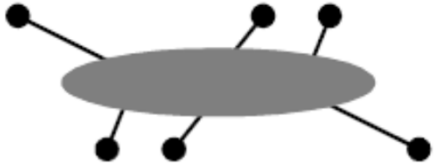


Common gaits with 6 legs



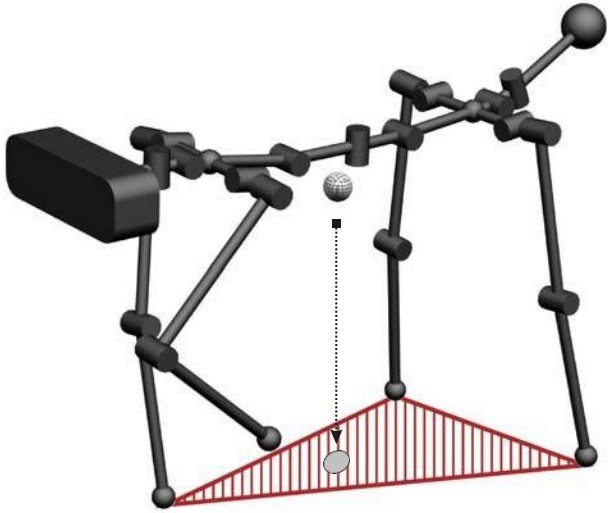






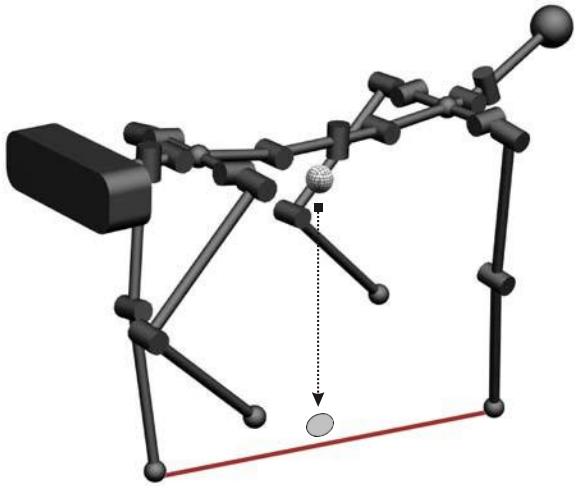
Different walking

- Statically stable



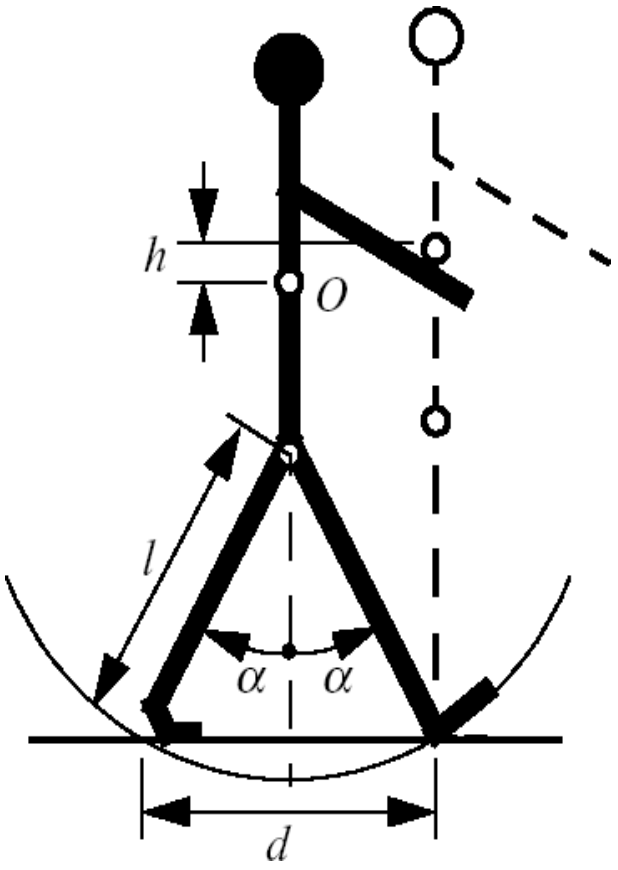
  - body weight supported by at least three legs
  - even if all joints ‘freeze’ instantaneously, the robot will not fall
  - safe, slow and inefficient

- Dynamic walking



  - the robot will fall if not continuously moving
  - less than three legs can be in ground contact
  - fast, efficient and demanding for actuation and control

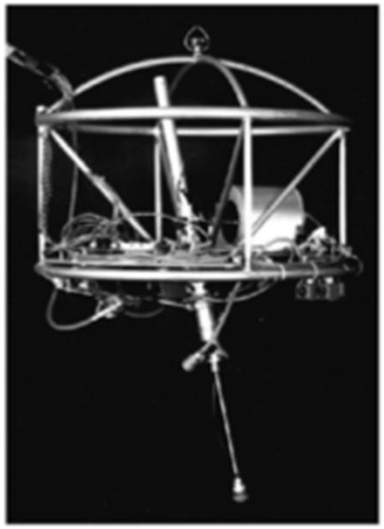
Biped walking

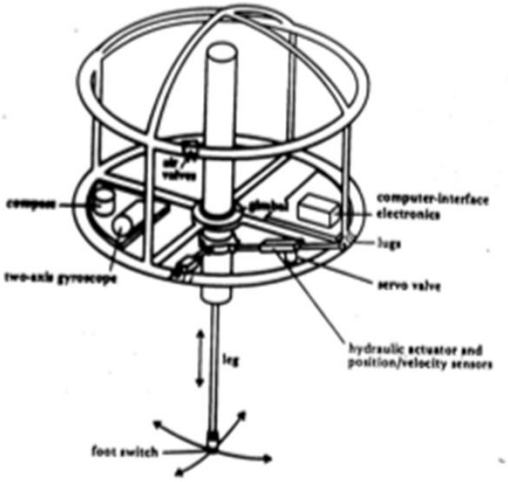


- Biped walking mechanism
  - not too far from real rolling
  - rolling of a polygon with side length equal to the length of the step
  - the smaller the step gets, the more the polygon tends to a circle (wheel)
- It is constrained in the facts
  - rotating joint was not invented by nature
  - work against gravity is required
  - more detailed analysis follows later in this presentation

Dynamic Locomotion

- Pneumatic piston
- Hydraulic leg “angle” orientation



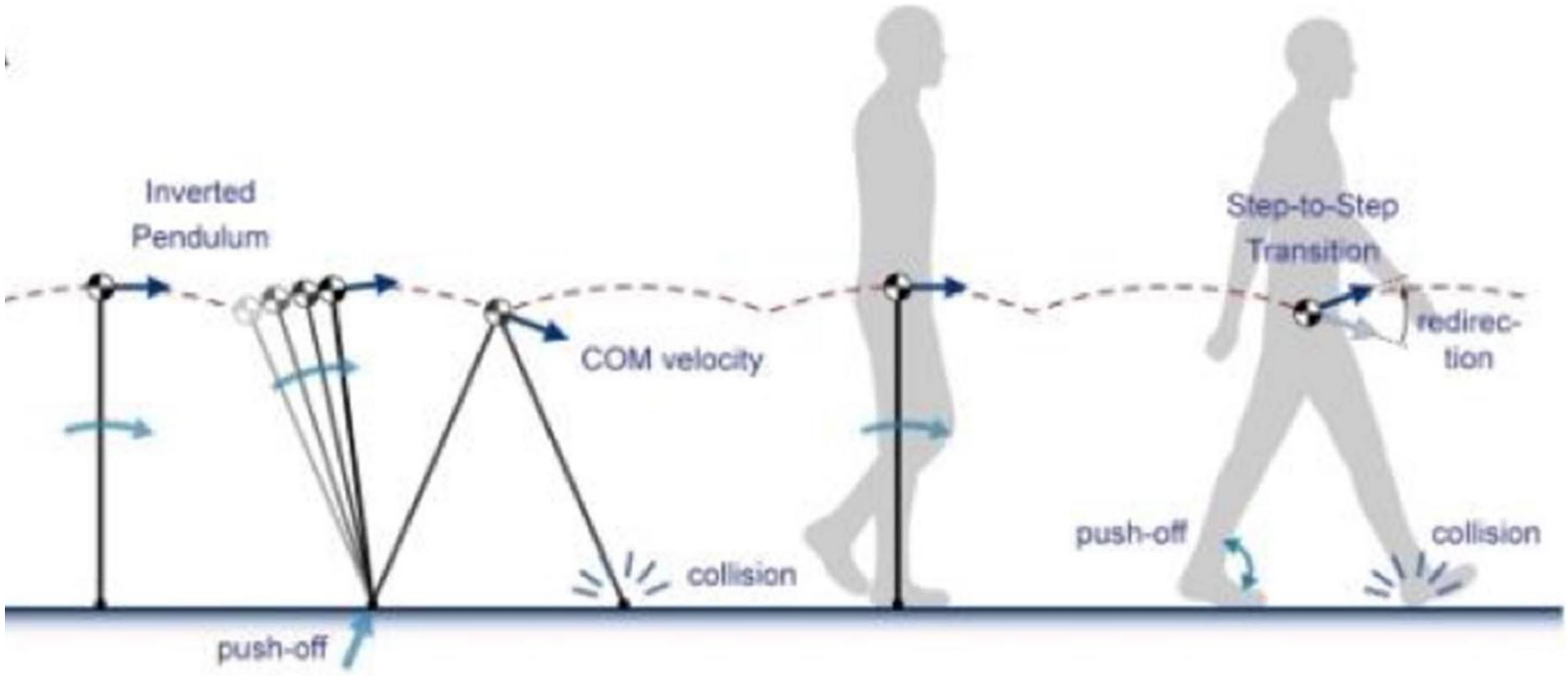


Early Raibert hoppers (MIT Leg Lab), 1983



Modelling of legged locomotion

- Walking can be represented by an inverted pendulum



Passive dynamic walker

- Forward falling combined with passive leg swing
- Storage of energy: potential ←→ kinetic in combination with low friction



Another case study

- Walking mechanism – Theo Jansen

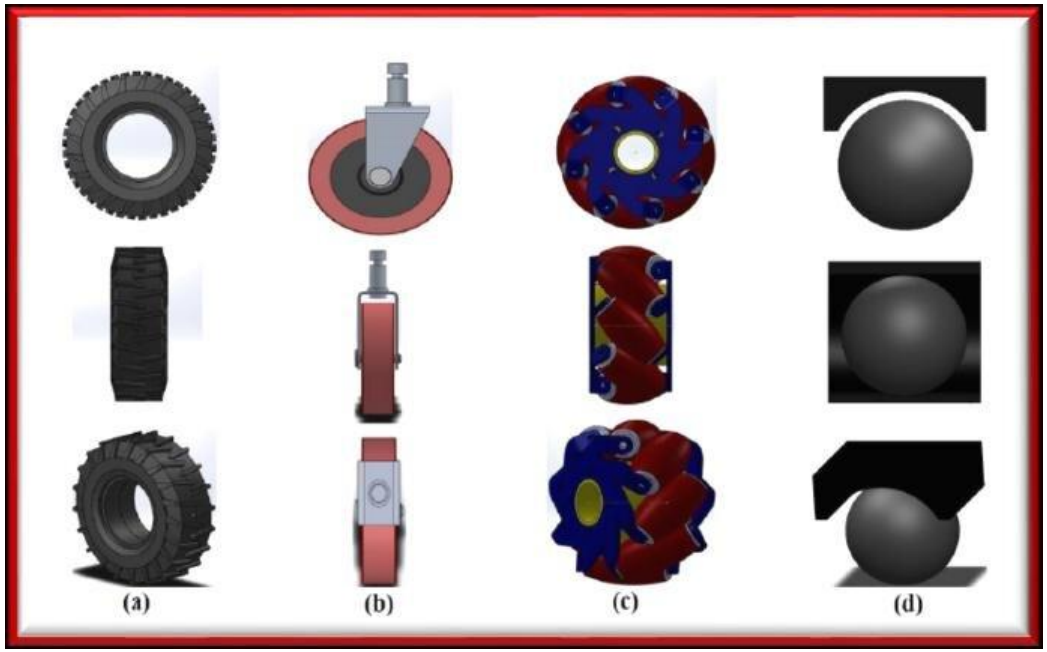


Mobile Wheeled Robots

- Wheels are the most appropriate solution for many applications
- Three wheels are sufficient to guarantee stability
- With more than three wheels an appropriate suspension is required
- Selection of wheels depends on the application

Wheel types for mobile mechanisms

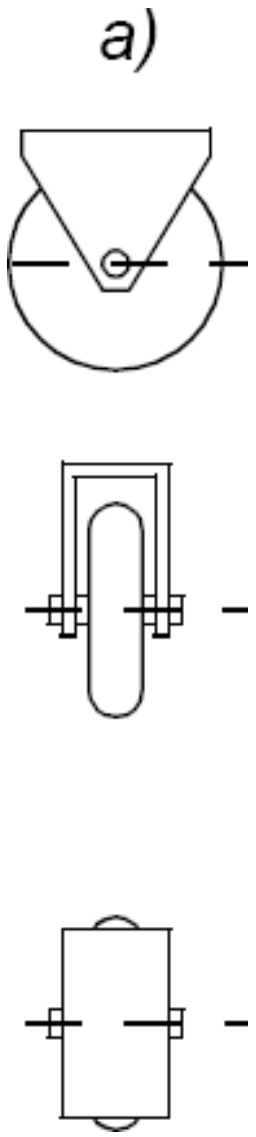
Standard, Caster, Swedish (Mecanum/Omni), and Spherical Wheels



- Why wheel choice matters: maneuverability, efficiency, load capacity, and control complexity
- Compare kinematics, pros/cons, and typical applications

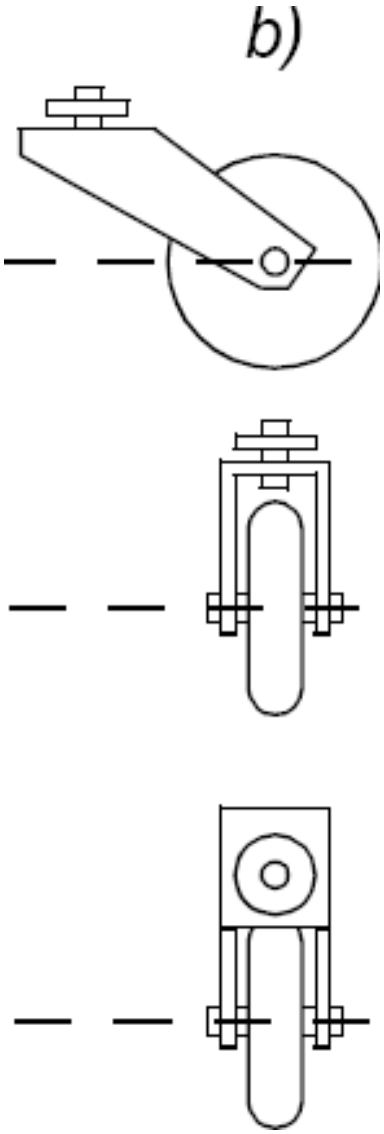
Standard wheel (fixed/steered)

- Definition: A rigid wheel with its rolling axis fixed relative to the chassis; steering may be provided by turning the entire wheel module.
- Pros: Highest energy efficiency & traction Simple mechanics, high load capacity, good at speed
- Cons: Limited maneuverability; larger turning radius; Requires **steering linkages** or differential control
- Typical applications: Cars, bicycles, carts, differential-drive robots



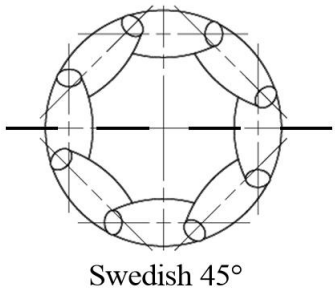
Caster wheel (swivel)

- Definition: A wheel mounted on a swivel fork that rotates about a vertical kingpin; the wheel axle is offset (trail).
- Pros: Very agile in confined spaces, Simple, inexpensive, widely available.
- Cons: Caster shimmy at high speeds; oscillation possible Scrub and wear during pivoting under load Lower precision and traction than driven wheels
- Typical applications: Trolleys, furniture, hospital beds, warehouse carts



45° Swedish wheel (Mecanum)

- Definition: A driven wheel with angled rollers (typically 45°) around the rim, allowing force components both forward and lateral.
- Pros: Holonomic-like maneuverability with wheels. Excellent in tight spaces.
- Cons: Lower traction and efficiency vs. standard wheels; Vibrations from roller transitions; sensitive to floor quality; Control requires careful kinematic mapping and motor coordination
- Typical applications: Warehouse robots, research platforms, stage automation





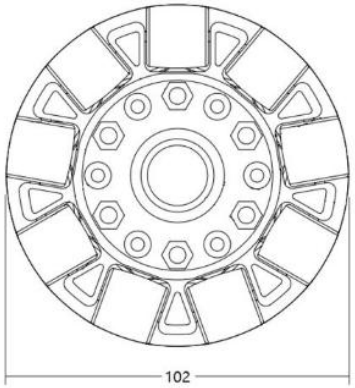
45° Swedish wheel (Mecanum)

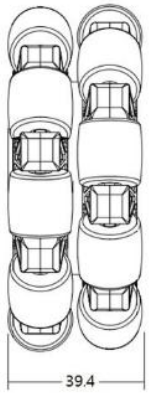


The Mecanum Wheel: How It Works

90° Swedish wheel (Omni wheel)

- Definition: A driven wheel with perpendicular free rollers around the rim; near-frictionless lateral motion, traction in the wheel's rolling direction.
- Pros: Omnidirectional motion with fewer vibrations than Mecanum; simple mechanical construction and widely available sizes.
- Cons: Reduced load capacity and traction vs. solid wheels; Floor smoothness matters; rollers can wear and accumulate debris
- Typical applications: Educational robots, small mobile platforms







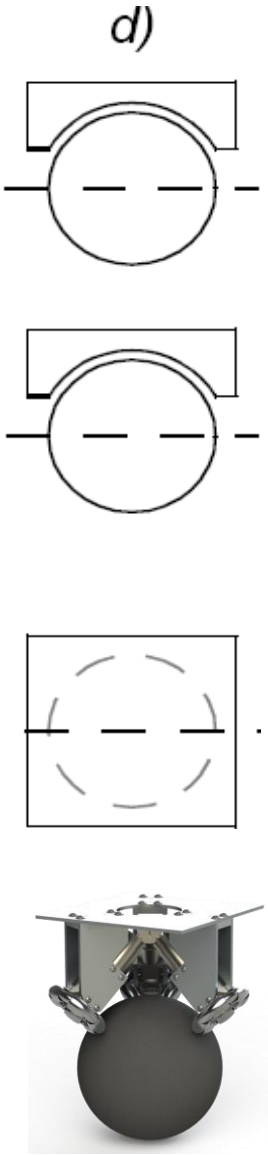
90° Swedish wheel (Omni wheel)



3WD Triangular 100mm omni wheel mobile robotics car C003

Spherical Wheel (ball wheel)

- Definition: A spherical element acting as the wheel; driven by multiple actuators/rollers contacting the ball to produce omnidirectional motion.
- Pros: Ultra-smooth omnidirectional motion; no wheel orientation changes Compact footprint; potentially excellent maneuverability.
- Cons: Complex drive/traction control and sealing Dirt ingress and lubrication issues; lower load capacity Implementation complexity and cost
- Typical applications: Ball transfer units (passive) and Ballbot concepts.



Comparative summary

- Maneuverability
  - Standard: ★★☆☆☆ (requires steering)
  - Caster: ★★☆☆☆ (great with push/pull, poor at speed)
  - Swedish (Mecanum/Omni): ★★★★★ (omnidirectional)
  - Spherical: ★★★★★ (omnidirectional, smooth)

- Efficiency / Traction
  - Standard: ★★★★★ (best)
  - Caster: ★★☆☆☆ (scrub & swivel losses)
  - Mecanum/Omni: ★★★☆☆ (roller losses)
  - Spherical: ★★☆☆☆ (multi-contact friction, control losses)

Comparative summary

- Load Capacity
  - Standard: ★★★★★ (best)
  - Caster: ★★★★☆
  - Mecanum/Omni: ★★★☆☆
  - Spherical: ★★☆☆☆
- Control Complexity
  - Standard: ★★☆☆☆ (steering/differential)
  - Caster: ★☆☆☆☆ (passive alignment)
  - Mecanum/Omni: ★★★★☆ (kinematic mapping)
  - Spherical: ★★★★★ (multi-actuator coordination)

Characteristics of Wheeled Robots

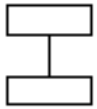
- The minimum number of wheels required for static stability is two.
- Stability is improved by 3 and more wheel. However, this arrangements are hyper static and require a flexible suspension system.





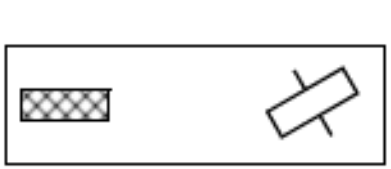
Cye, a domestic robot, was designed to vacuum floors and make domestic deliveries. A two-wheel differential-drive robot can achieve static stability if the center of mass is below the wheel axle.

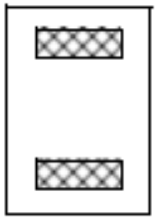
Icons for each wheel type

	unpowered omnidirectional wheel (spherical, castor, Swedish)
	motorized Swedish wheel (Stanford wheel)
	unpowered standard wheel
	motorized standard wheel
	motorized and steered castor wheel
	steered standard wheel
	connected wheels

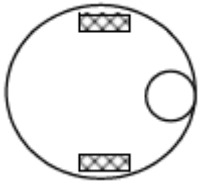
Different Arrangements of Wheels

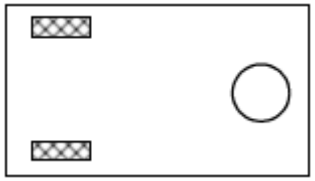
- Two wheels

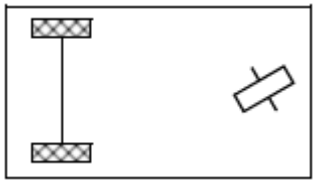


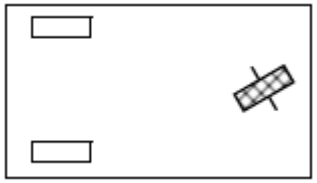


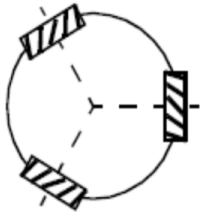
- Three wheels



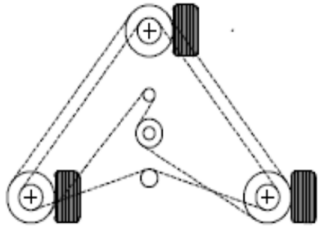








Omnidirectional Drive

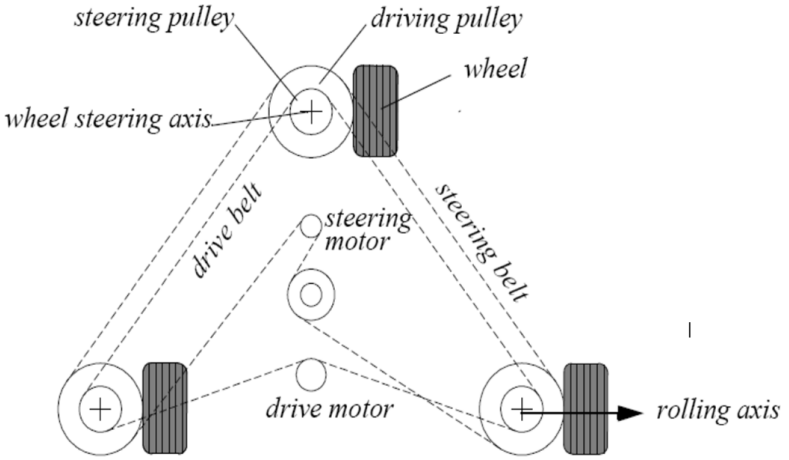


Synchro Drive

Synchro Drive

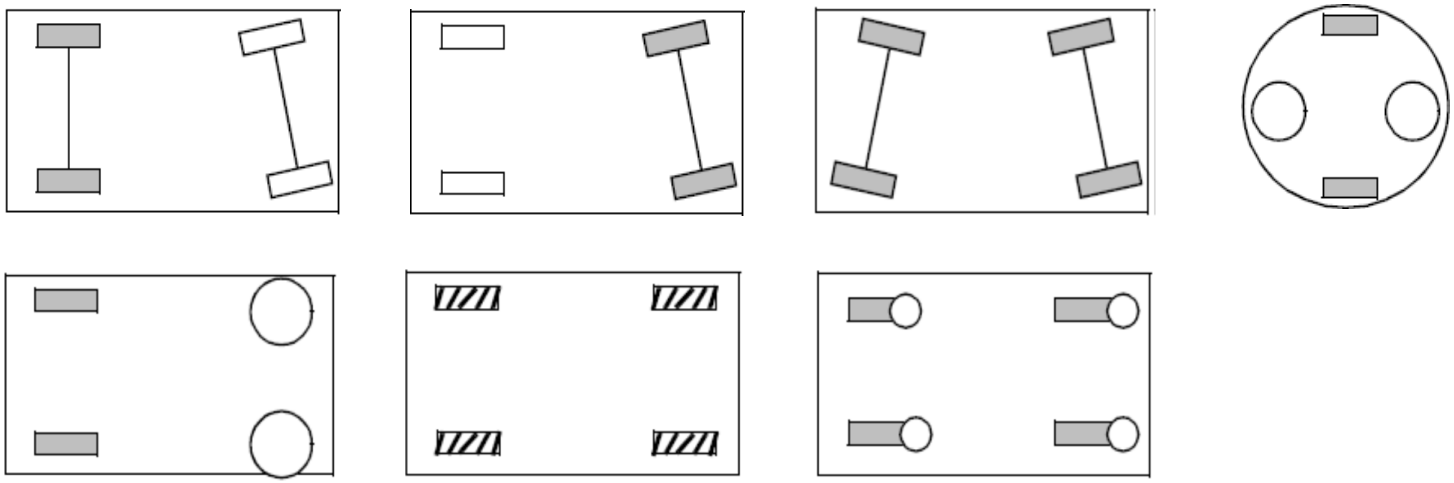
- All wheels are actuated synchronously by one motor: defines the speed of the vehicle
- All wheels steered synchronously by a second motor: sets the heading of the vehicle
- The orientation in space of the robot frame will always remain the same. It is therefore not possible to control the orientation of the robot frame.



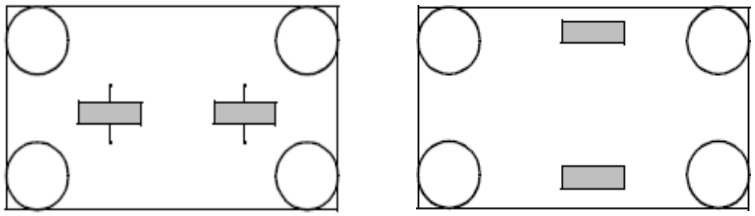


Different Arrangements of Wheels

- Four wheels

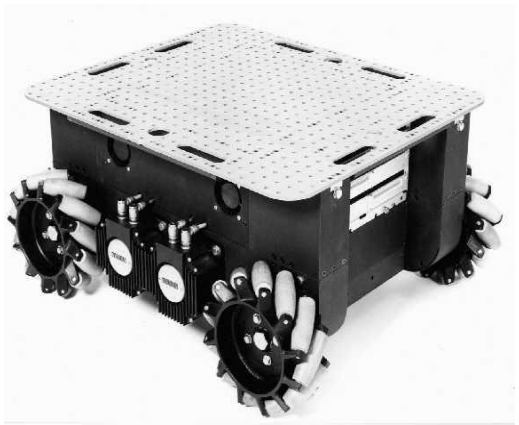


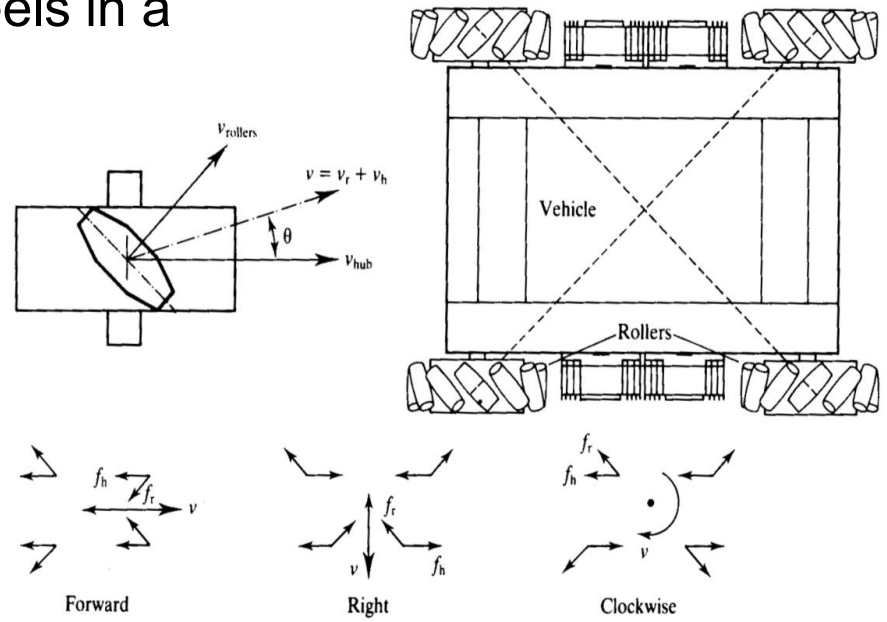
- Six wheels



CMU Uranus

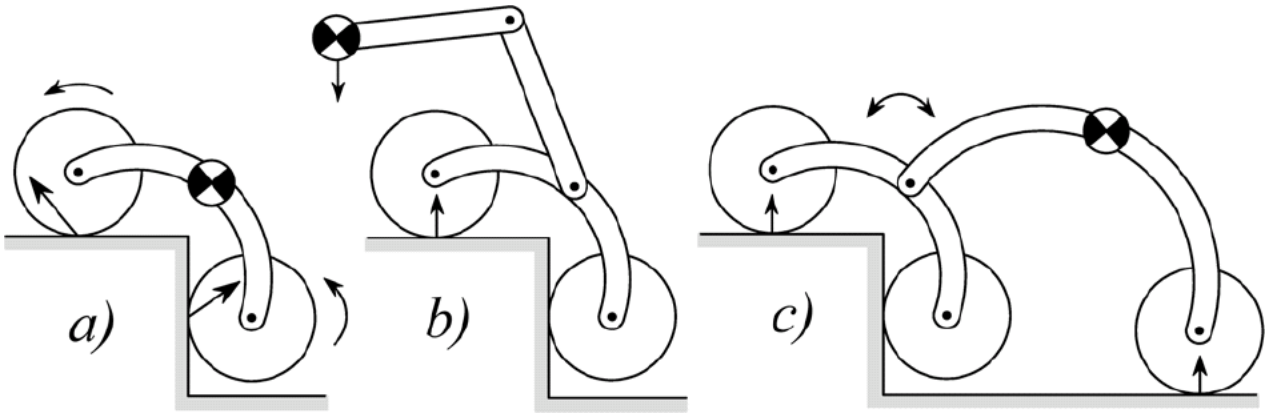
- Omnidirectional Drive with 4 Wheels
- Movement in the plane has 3 DOF
  - thus only three wheels can be independently controlled
  - It arranges three Swedish wheels in a triangle





Wheeled Rovers

- Concepts for Object Climbing: Purely friction based



Change of center of gravity (CoG)

Adapted suspension mechanism with passive or active joints

Climbing with Legs: Shrimp

- Passive locomotion concept
- 6 wheels
  - two boogies on each side
  - fixed wheel in the rear
  - front wheel with spring suspension
- Dimensions
  - length: 60 cm
  - height: 20 cm
- Characteristics
  - highly stable in rough terrain
  - overcomes obstacles up to 2 times its wheel diameter



ROSE5730 Autonomous Mobile Robots

Lecture 2: Locomotion

Dongkun Han
Department of Mechanical and Automation Engineering;
The Chinese University of Hong Kong.

